$$\frac{1}{C_{eq}} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \qquad \Rightarrow C_{eq} = 2\ \mu F$$

and

$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \qquad \Rightarrow C_{eq} = 4\ \mu F$$

From the figure (c), we infer that 2 μF and 4 μF are connected in parallel. So the equivalent capacitance is given in the figure (d).

$$C_{eq} = 2 + 4 = 6\ \mu F$$

Thus the combination of capacitances in figure (a) can be replaced by a single capacitance 6 μF.

## 1.9 DISTRIBUTION OF CHARGES IN A CONDUCTOR AND ACTION AT POINTS

### 1.9.1 Distribution of charges in a conductor

Consider two conducting spheres A and B of radii $r_1$ and $r_2$ respectively connected to each other by a thin conducting wire as shown in the Figure 1.60. The distance between the spheres is much greater than the radii of either spheres.

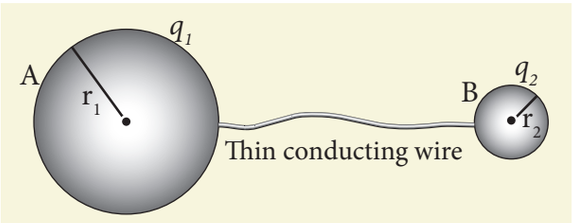



**Figure 1.60** Two conductors are connected through conducting wire

If a charge $Q$ is introduced into any one of the spheres, this charge $Q$ is redistributed into both the spheres such that the electrostatic potential is same in both the spheres. They are now uniformly charged and attain electrostatic equilibrium. Let $q_1$ be the charge residing on the surface of sphere A and $q_2$ is the charge residing on the surface of sphere B such that $Q = q_1 + q_2$. The charges are distributed only on the surface and there is no net charge inside the conductor.

The electrostatic potential at the surface of the sphere A is given by

$$V_A = \frac{1}{4\pi\epsilon_\circ}\frac{q_1}{r_1} \qquad (1.109)$$

The electrostatic potential at the surface of the sphere B is given by

$$V_B = \frac{1}{4\pi\epsilon_\circ}\frac{q_2}{r_2} \qquad (1.110)$$

The surface of the conductor is an equipotential. Since the spheres are connected by the conducting wire, the surfaces of both the spheres together form an equipotential surface. This implies that

$$V_A = V_B$$
$$\text{or} \quad \frac{q_1}{r_1} = \frac{q_2}{r_2} \qquad (1.111)$$

Let the charge density on the surface of sphere A be $\sigma_1$ and that on the surface of sphere B be $\sigma_2$. This implies that $q_1 = 4\pi r_1^2\sigma_1$ and

$q_2 = 4\pi r_2^2\sigma_2$. Substituting these values into equation (1.112), we get

$$\sigma_1 r_1 = \sigma_2 r_2 \qquad (1.112)$$

from which we conclude that

$$\sigma r = \text{constant} \qquad (1.113)$$



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